BILLY & BUDDY SERVICE DESIGN

A mini game that teaches kids to tell a need from a want, in two minutes

ROLE	METHODS	TYPE	TEAM
Website mini games. Story and narrative, with Kaitlyn Carney.	Game ideation, physical prototype test, service mapping	Student project, COllab Studio	Five designers. Visual language and illustration by Sharanya.

TL;DR

Only about half of American schools require a class on money. A child's chance of learning how saving, spending and budgeting work comes down to which state they live in, and the gap does not close on its own.

Billy & Buddy is a sixteen-book series delivered a book at a time, with mini games on the website that replay each book's lesson. I designed the game for needs against wants. A child picks a situation, sorts ten items into two baskets on a two-minute clock, then spends a fixed budget at the checkout. Billy answers every choice as it happens.

RESEARCH	DESIGN	STATUS
Team research on financial literacy education, done before I joined. One physical prototype test of the game logic.	Game concept, two-part structure, full state map, website flow, and four service artifacts.	Designed and prototyped on paper. The digital version has not been tested with children.

BRIEF

Teach American children ages five to nine about money through books, and reinforce each lesson on a digital platform.

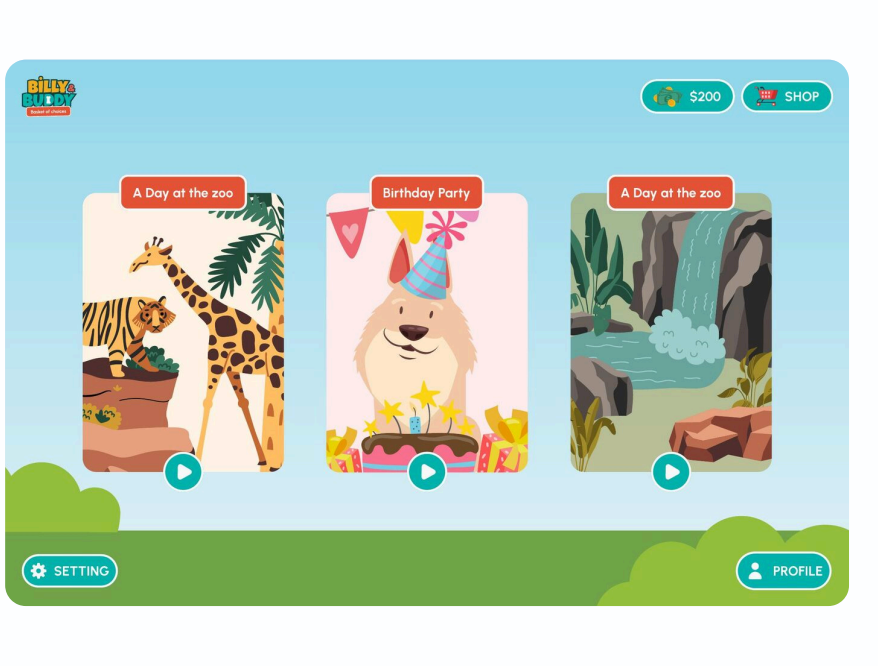
Each book carries one age-appropriate concept in an interactive story. The website is where the concept gets used again, in a game short enough that a child will actually finish it.

CHALLENGE

How can a six-year-old learn to tell a need from a want in the two minutes they will actually give a game?

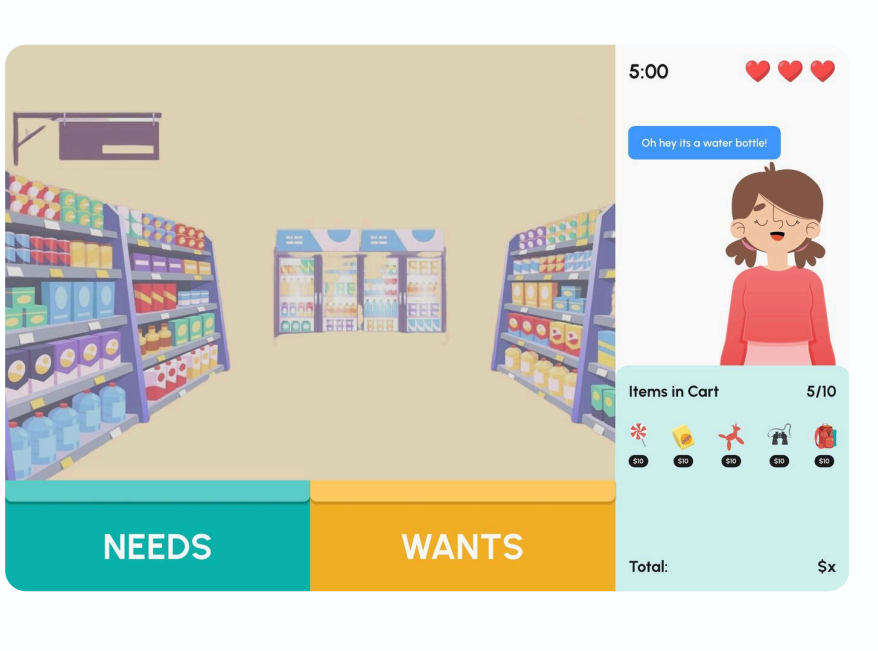
SOLUTION

Four screens carry the idea. Here is what each one does.



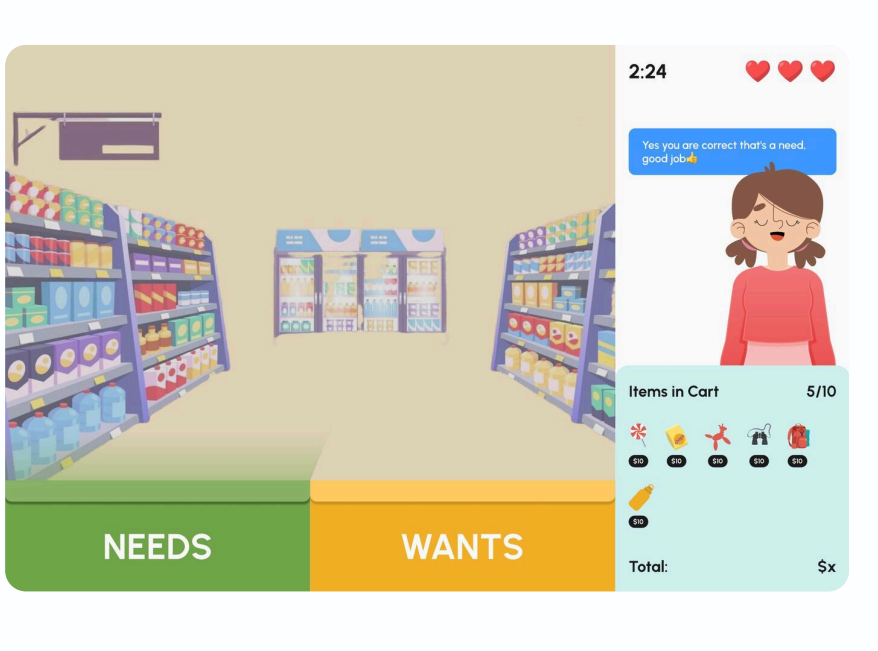
The lesson starts with a situation, not a definition.

A player picks a context first. A day at the zoo, a birthday party. The same item can be a need in one situation and a want in another, so the game hands over the situation before it asks anything.



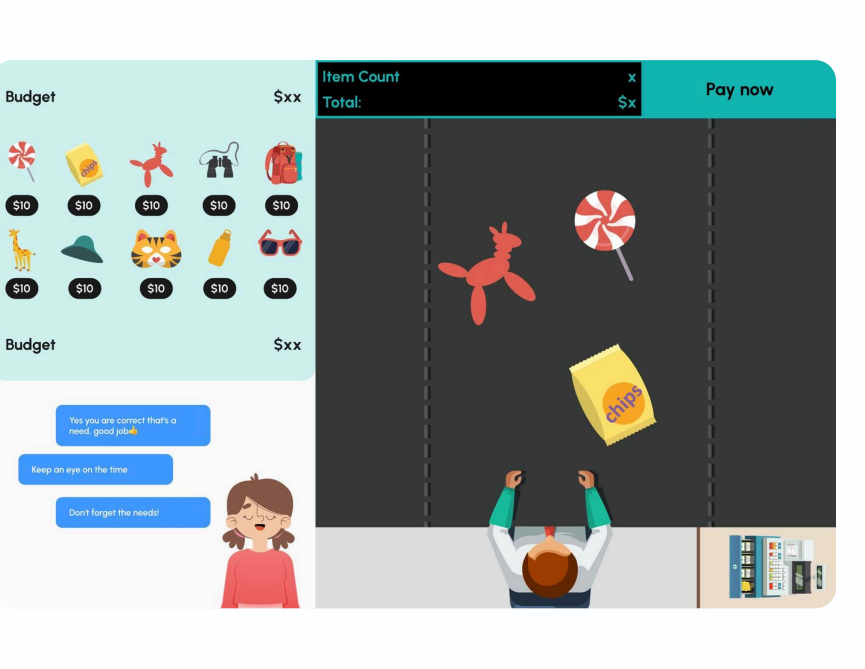
Ten items, two baskets, two minutes.

The player sorts ten items into Needs and Wants against a clock, with three lives. A wrong basket costs one. The clock is what forces a decision instead of a lookup.



Billy answers every choice out loud.

Right or wrong, the response lands in the same second as the tap. A wrong sort says so and offers the next try rather than ending the round.



Sorting is half of it. Then you have to pay.

Part two hands the player a budget. Every need has to make it into the cart, wants only fit if there is room, and money left over adds to the score. Knowing the difference is not the same as spending on it.

THE PROBLEM

Financial literacy for children fails in four ordinary ways.

Half of schools do not teach it at all.

In American primary and secondary education, only about half of schools require classes on financial literacy. As of 2024, 26 states required it. Whether a child learns this comes down to an address.

Four in seven adults can be considered financially literate.

The gap does not close with age. Children who never learn this become adults who cannot make informed decisions about their own money.

The resources that exist hand over information, not practice.

There are services that give students the basics. Few of them teach money gradually, in a form a child will sit through more than once.

A definition does not survive contact with a supermarket aisle.

A child can repeat what a need is and still put candy in the basket. The concept has to be used inside a situation or it stays a word.

INITIAL FINDINGS

Five findings from the team's research, and the two that decided the format.

> I joined this project after the research was done. The five findings below are the team's work. I am reporting them, not claiming them.

Learning sticks when it looks like the situation it is for.

Real-life scenarios bridge the gap between theory and practice.

Attention is the constraint, not comprehension.

Interactive and gamified learning with incentives is the most effective way to hold a child's attention.

Small pieces beat one long explanation.

Structuring learning in chunks is the most accessible format and the most likely to be retained.

Nobody agrees what a child should know by adulthood.

There is no current standard for a child's financial literacy on entering the teenage years or adult life.

The gap compounds.

Financially illiterate children grow into financially illiterate adults who cannot make smart and informed financial decisions.

Findings two and three point at the same thing. If attention is the limit and small chunks are the unit, the reinforcement has to be short, situated and repeatable. That is a game, not a chapter.

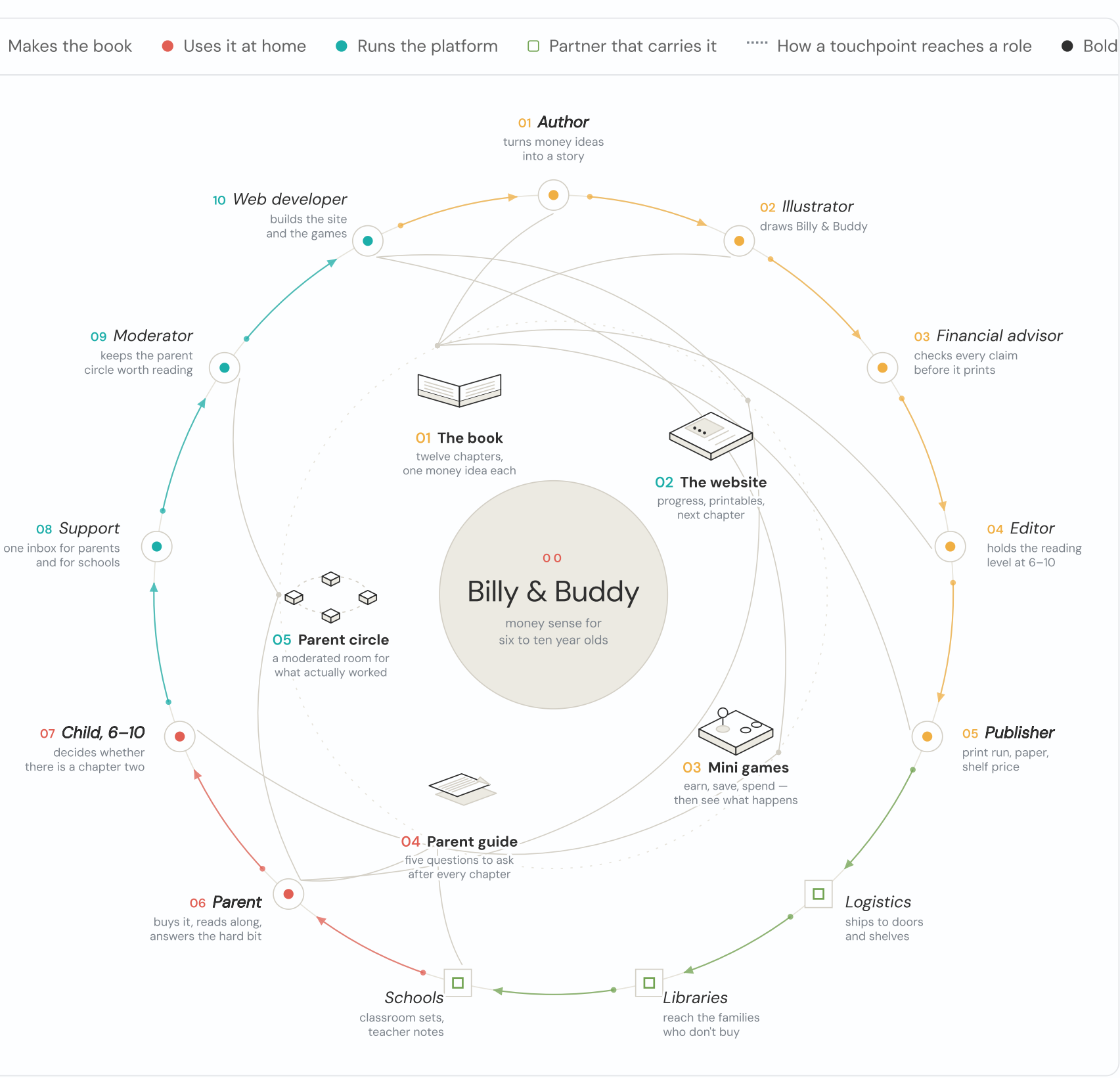
THE SYSTEM

Before designing the game I mapped what it sits inside.

A game that reinforces a book is only worth building if the book arrives on a schedule, the site is where a child goes between deliveries, and somebody is paying for both. These four artifacts describe that. They are redrawn here from the process book, with the content unchanged.

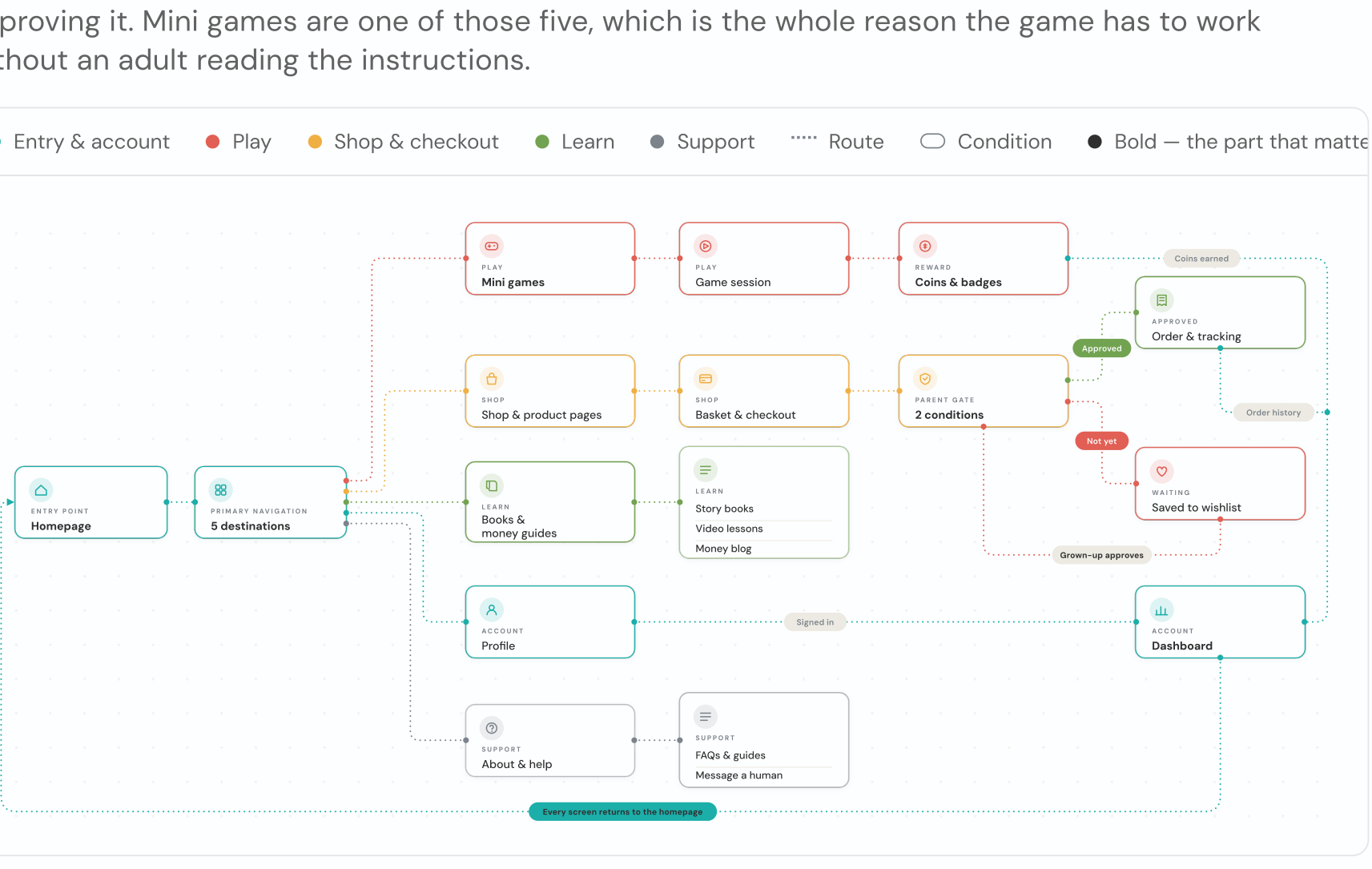
Stakeholder map. Ten people, three partners, and five things a family ever actually touches.

The arrows run clockwise: five people make the book, three partners carry it out into the world, a parent and a child use it, and what the site learns on the way round comes back to the author. Numbering follows that loop rather than importance, and colour says which side someone is on — amber makes the book, coral uses it at home, teal runs the platform.


Makes the book
Uses it at home
Runs the platform
Partner that carries it
How a touchpoint reaches a role
Bold
01 Author
turns money ideas into a story
02 Illustrator
draws Billy & Buddy
03 Financial advisor
checks every claim before it prints
04 Editor
holds the reading level at 6–10
05 Publisher
print run, paper, shelf price
Logistics
ships to doors and shelves
Libraries
reach the families who don't buy
Schools
classroom sets, teacher notes
06 Parent
buys it, reads along, answers the hard bit
07 Child, 6–10
decides whether there is a chapter two
08 Support
one inbox for parents and for schools
09 Moderator
keeps the parent circle worth reading
10 Web developer
builds the site and the games
01 The book
twelve chapters, one money idea each
02 The website
progress, printables, next chapter
03 Mini games
earn, save, spend — then see what happens
04 Parent guide
five questions to ask after every chapter
05 Parent circle
a moderated room for what actually worked
00
Billy & Buddy
money sense for six to ten year olds

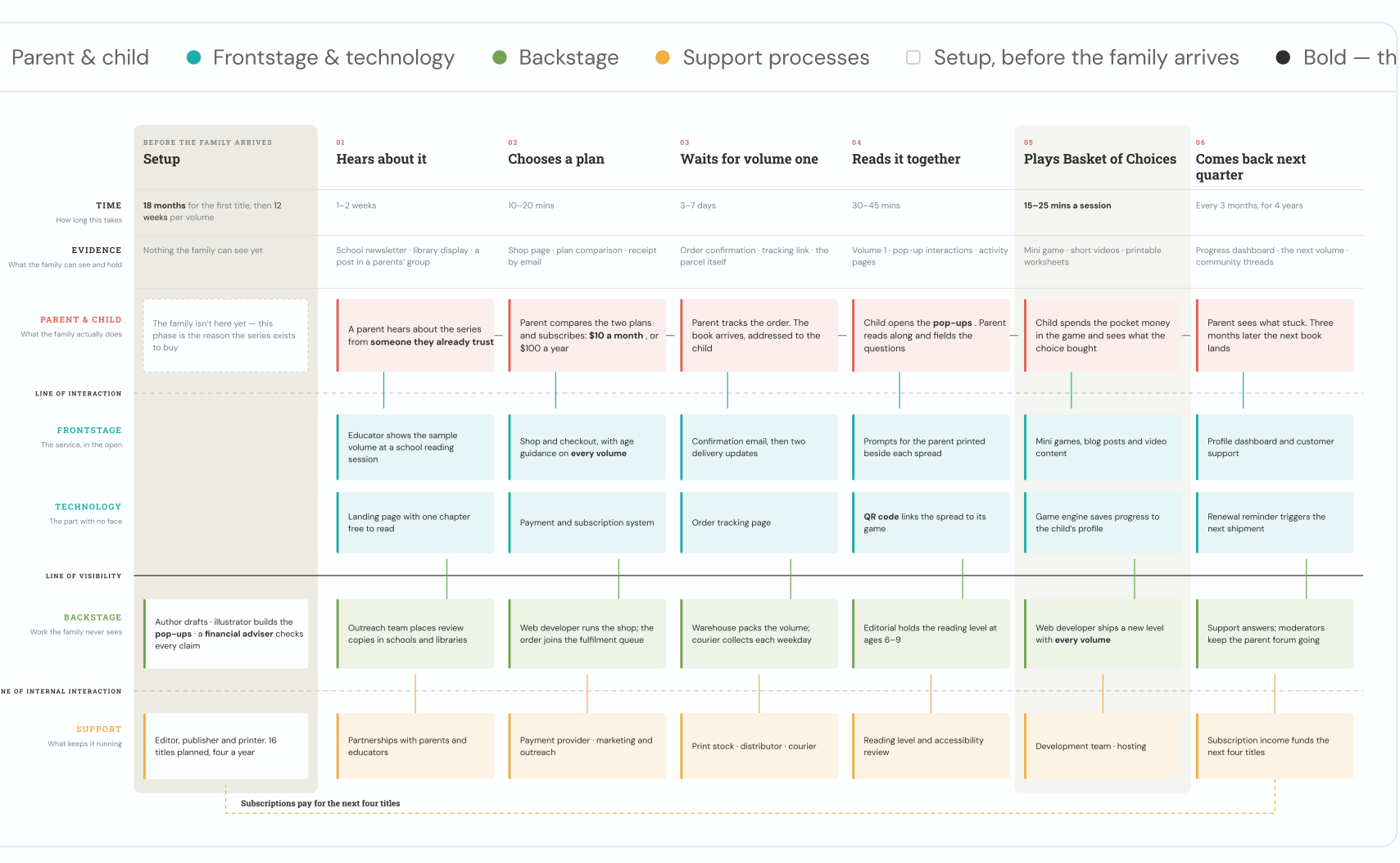
System diagram. Every route through the website, and the one gate that stops a child buying alone.

The homepage opens onto five destinations, and every screen carries a route back to it. Play and shopping both feed the same profile dashboard, and checkout cannot complete without a grown-up approving it. Mini games are one of those five, which is the whole reason the game has to work without an adult reading the instructions.



Service blueprint. One setup phase, then six stages of the family's journey.

Five lanes, from a blank page to a child playing. The family does not appear until stage one; everything in the setup column is people the family never sees. Eighteen months for the first title, then twelve weeks per volume, and the subscriptions from stage six pay for the next four titles.

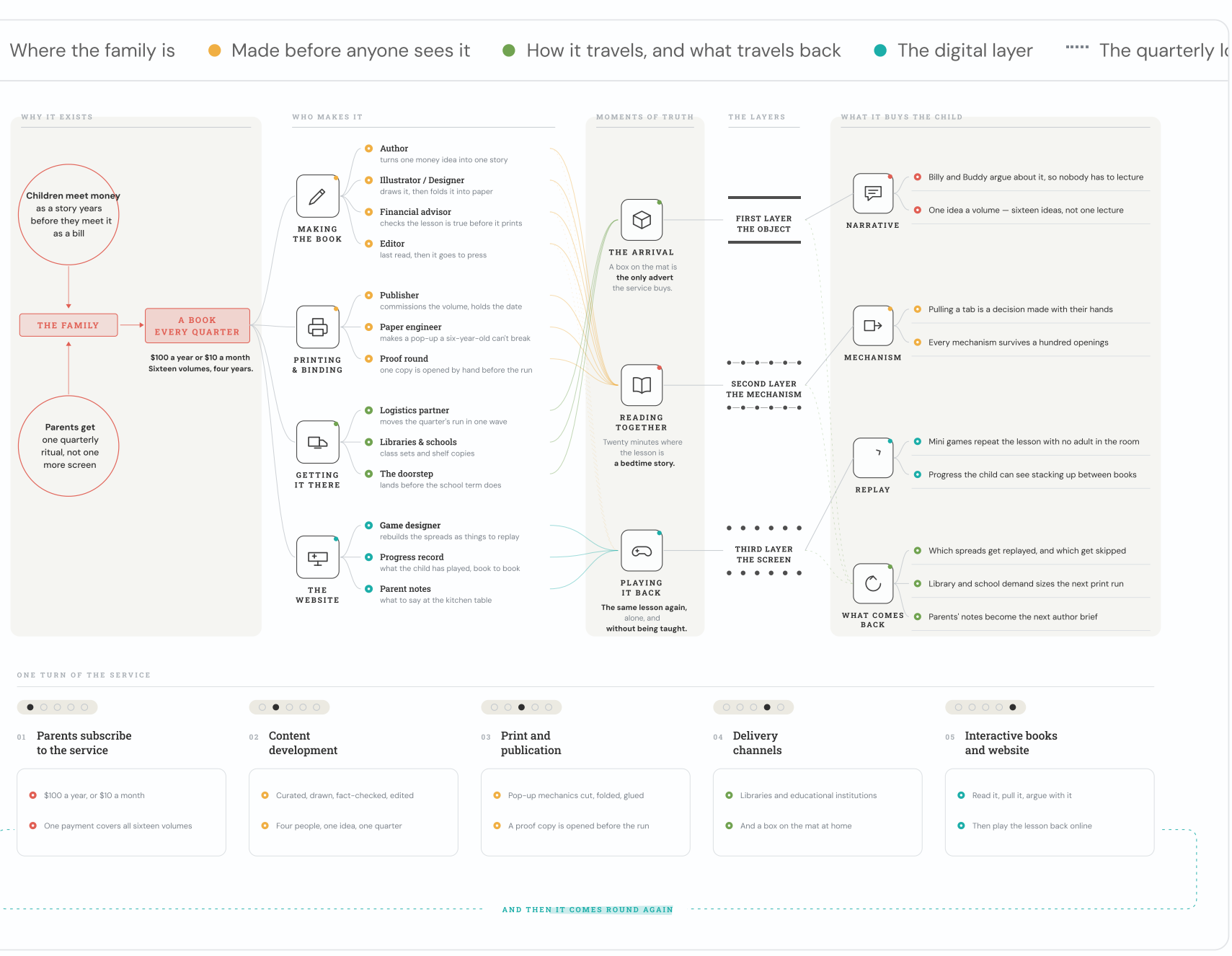

Parent & child
Frontstage & technology
Backstage
Support processes
Setup, before the family arrives
Bold — th

	BEFORE THE FAMILY ARRIVES Setup	01 Hears about it	02 Chooses a plan	03 Waits for volume one	04 Reads it together	05 Plays Basket of Choices	06 Comes back next quarter
TIME How long this takes	**18 months** for the first title, then 12 weeks per volume	1–2 weeks	10–20 mins	3–7 days	30–45 mins	**15–25 mins a session**	Every 3 months, for 4 years
EVIDENCE What the family can see and hold	Nothing the family can see yet	School newsletter · library display · a post in a parents' group	Shop page · plan comparison · receipt by email	Order confirmation · tracking link · the parcel itself	Volume 1 · pop-up interactions · activity pages	Mini game · short videos · printable worksheets	Progress dashboard · the next volume · community threads
PARENT & CHILD What the family actually does	The family isn't here yet — this phase is the reason the series exists to buy	A parent hears about the series from **someone they already trust**	Parent compares the two plans and subscribes: **$10 a month**, or $100 a year	Parent tracks the order. The book arrives, addressed to the child	Child opens the **pop-ups**. Parent reads along and fields the questions	Child spends the pocket money in the game and sees what the choice bought	Parent sees what stuck. Three months later the next book lands
LINE OF INTERACTION							
FRONTSTAGE The service, in the open		Educator shows the sample volume at a school reading session	Shop and checkout, with age guidance on **every volume**	Confirmation email, then two delivery updates	Prompts for the parent printed beside each spread	Mini games, blog posts and video content	Profile dashboard and customer support
TECHNOLOGY The part with no face		Landing page with one chapter free to read	Payment and subscription system	Order tracking page	**QR code** links the spread to its game	Game engine saves progress to the child's profile	Renewal reminder triggers the next shipment
LINE OF VISIBILITY							
BACKSTAGE Work the family never sees	Author drafts · illustrator builds the **pop-ups** · a **financial adviser** checks every claim	Outreach team places review copies in schools and libraries	Web developer runs the shop; the order joins the fulfilment queue	Warehouse packs the volume; courier collects each weekday	Editorial holds the reading level at ages 6–9	Web developer ships a new level with **every volume**	Support answers; moderators keep the parent forum going
LINE OF INTERNAL INTERACTION							
SUPPORT What keeps it running	Editor, publisher and printer. 16 titles planned, four a year	Partnerships with parents and educators	Payment provider · marketing and outreach	Print stock · distributor · courier	Reading level and accessibility review	Development team · hosting	Subscription income funds the next four titles

Subscriptions pay for the next four titles

Service visualization. One turn of the service, then the loop.

A parent signs up, four people make a book, a courier carries it, and a child plays the lesson back. Sixteen volumes across four years at $100 a year or $10 a month. A book arrives every quarter, which is the gap the game has to bridge.


Where the family is
Made before anyone sees it
How it travels, and what travels back
The digital layer
The quarterly l
WHY IT EXISTS
Children meet money as a story years before they meet it as a bill
THE FAMILY
A BOOK EVERY QUARTER
$100 a year or $10 a month
Sixteen volumes, four years.
Parents get one quarterly ritual, not one more screen
WHO MAKES IT
MAKING THE BOOK
Author
turns one money idea into one story
Illustrator / Designer
draws it, then folds it into paper
Financial advisor
checks the lesson is true before it prints
Editor
last read, then it goes to press
PRINTING & BINDING
Publisher
commissions the volume, holds the date
Paper engineer
makes a pop-up a six-year-old can't break
Proof round
one copy is opened by hand before the run
GETTING IT THERE
Logistics partner
moves the quarter's run in one wave
Libraries & schools
class sets and shelf copies
The doorstep
lands before the school term does
THE WEBSITE
Game designer
rebuilds the spreads as things to replay
Progress record
what the child has played, book to book
Parent notes
what to say at the kitchen table
MOMENTS OF TRUTH
THE ARRIVAL
A box on the mat is the only advert the service buys.
READING TOGETHER
Twenty minutes where the lesson is a bedtime story.
PLAYING IT BACK
The same lesson again, alone, and without being taught.
THE LAYERS
FIRST LAYER THE OBJECT
SECOND LAYER THE MECHANISM
THIRD LAYER THE SCREEN
WHAT IT BUYS THE CHILD
NARRATIVE
Billy and Buddy argue about it, so nobody has to lecture
One idea a volume — sixteen ideas, not one lecture
MECHANISM
Pulling a tab is a decision made with their hands
Every mechanism survives a hundred openings
REPLAY
Mini games repeat the lesson with no adult in the room
Progress the child can see stacking up between books
WHAT COMES BACK
Which spreads get replayed, and which get skipped
Library and school demand sizes the next print run
Parents' notes become the next author brief
ONE TURN OF THE SERVICE
01 Parents subscribe to the service
$100 a year, or $10 a month
One payment covers all sixteen volumes
02 Content development
Curated, drawn, fact-checked, edited
Four people, one idea, one quarter
03 Print and publication
Pop-up mechanics cut, folded, glued
A proof copy is opened before the run
04 Delivery channels
Libraries and educational institutions
And a box on the mat at home
05 Interactive books and website
Read it, pull it, argue with it
Then play the lesson back online
AND THEN IT COMES ROUND AGAIN

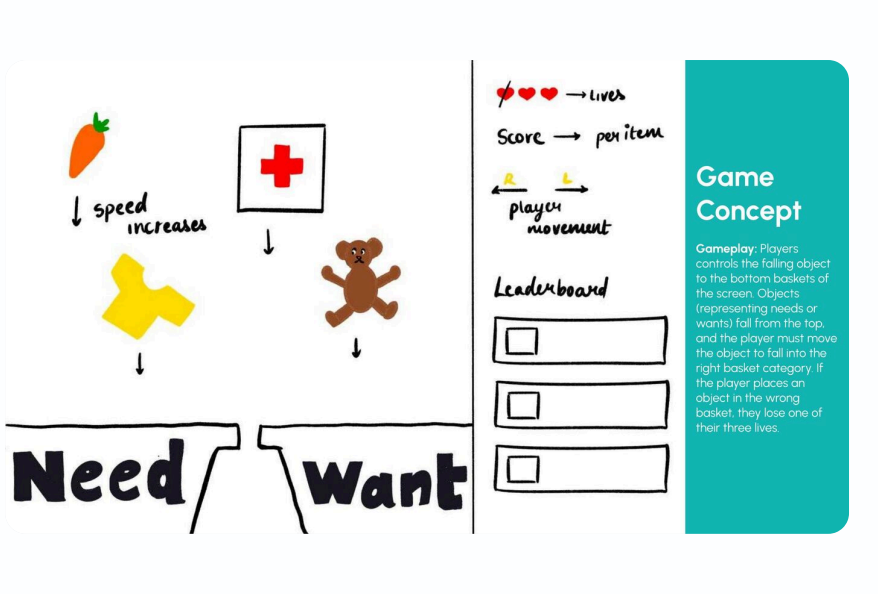
DESIGN PROCESS

Grid first, then a game, then paper.

Testing out game logic by adding complexities, changing goals, and adding parameters

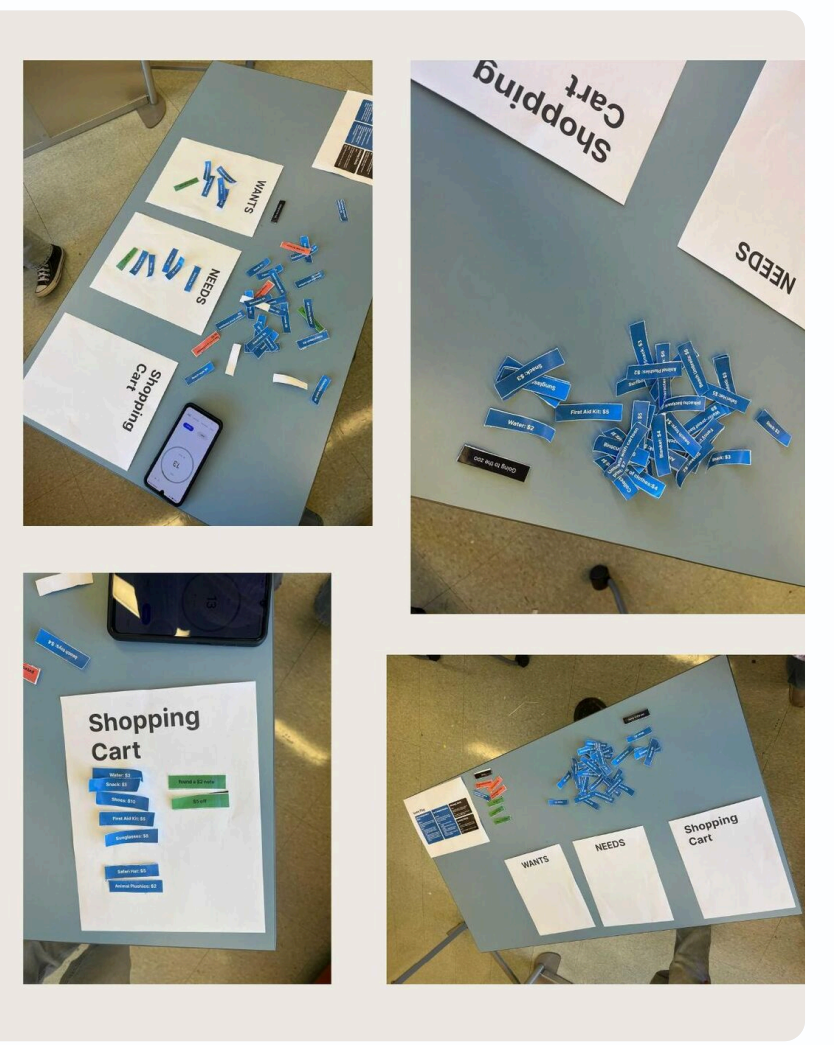
I tested the grid before I tested the game.

The first round was structure. How a grid changes difficulty, what happens when the goal moves, what each new parameter costs in confusion. Cheap to draw and quick to throw away.



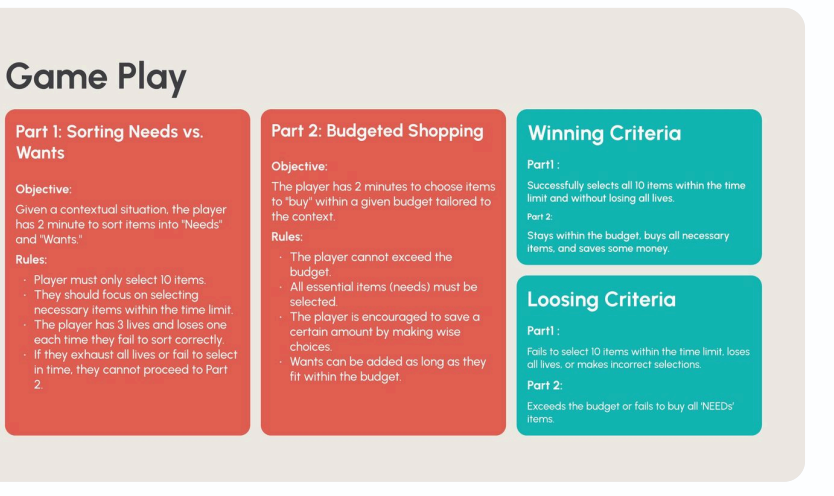
The first working idea was a falling-object game.

Objects drop from the top and the player steers them into Needs or Wants. Speed rises each level. Three lives, bonuses, sale coupons. It was fun to describe and it did not survive the test.



Then I built it out of paper and watched it fail.

I tested the logic as a physical prototype with a shopping cart, item cards and a time limit. Three things came back. One of them ended the falling-object version.


Game Play
Part 1: Sorting Needs vs. Wants
Part 2: Budgeted Shopping
Winning Criteria
Loosing Criteria

The rules that survived.

Part one is sorting, two minutes, ten items, three lives. Part two is a budget at the checkout where every need is mandatory. Failing part one blocks part two, so the sorting has to be learned before the spending.

TURNING POINT

The paper test ended the falling-object version.

FINDING ONE

The predetermined categories for Needs and Wants did not align with players' own view of what counts as a need. That caused confusion and mismatches during play.

FINDING TWO

The dog character was not effectively integrated into gameplay, so it felt disconnected and secondary, despite being intended as a companion and guide.

FINDING THREE

Physical play with a time constraint did not capture the digital experience the game was for. It said nothing about how a child would handle a UI, real-time feedback or pacing.

WHAT I CHANGED

The falling-object mechanic went. Sorting became deliberate and timed instead of reflex-driven, a second part was added so the concept has to be spent as well as named, and the guide now speaks on every single action.

> The first finding is still open. The fix is a survey of children in the target age range on which items they call a need, and then rebuilding the item list from their answers rather than mine. I have not run it.

GOAL

Three things the game has to prove, none of them proven yet.

A child should be able to sort without asking an adult. UNTESTED

Categories have to be clear enough, and items obvious enough, for an independent decision inside the time limit.

The budget should teach itself. UNTESTED

A player has to see the total move as items go in and out, and work out on their own that the needs are not optional.

The trade-off is the actual lesson. UNTESTED

What matters is whether a child reasons about leftover money, and whether overspending on wants registers as a consequence rather than bad luck.

THE BUSINESS CASE

Who pays for this, and the part I could not resolve.

THE MODEL

A subscription at $100 a year or $10 a month delivers one book at a time across sixteen books and four years, with revenue also from single book sales and branded merchandise. The buyers are parents who want their child financially literate and educators who want something they can drop into a classroom without building it.

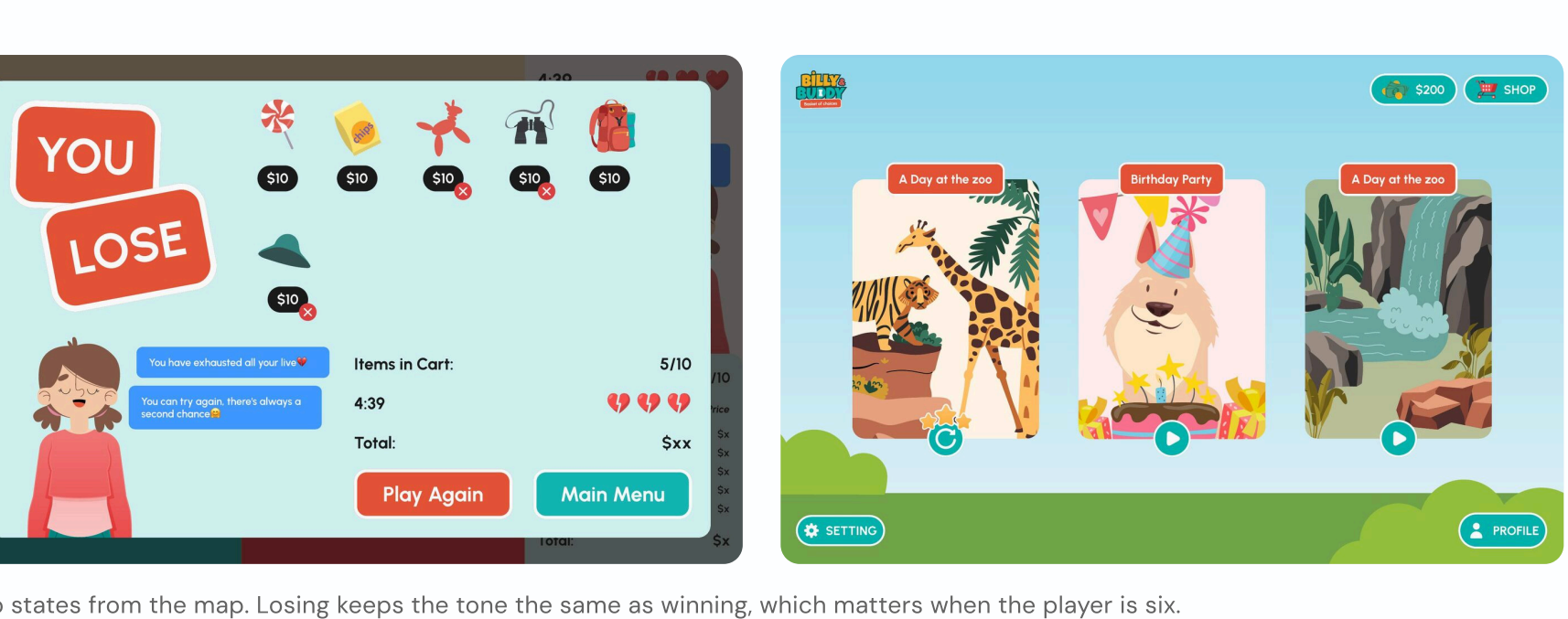
The game is free, and the lesson inside it is to want less. MY READ

Its job is to bring children back between deliveries. The service earns more when families stay subscribed and buy more, while the game teaches spending less. Naming that is not solving it, and I did not solve it.

OUTCOME

What exists, and the test that would settle it.

A full game concept with a two-part structure and a mapped state set: three lose states, two win states, a checkout, and victory scoring. Alongside it, the website flow and four service artifacts. The physical prototype tested the logic. Nothing has been tested with children, and nothing has been tested on a screen.



Two states from the map. Losing keeps the tone the same as winning, which matters when the player is six.

The test I would run.

A clickable prototype, timed, with children in the target age range. Three things to watch: whether they sort without asking, whether the budget total makes sense to them as it moves, and whether any of them choose to leave money unspent. Before that, the survey on what a child calls a need. The paper test already told me the item list is the weak point.

REFLECTION

WHAT I TAKE FORWARD

Test the medium, not only the logic.

Paper told me the rules worked and nothing about the timer, which was the whole point of the design. The prototype has to be made of the same material as the thing.

The categories have to come from the player.

I wrote the item list from an adult's idea of a need. The first test found that in minutes, and it is the finding I have not yet answered.

A companion character needs a job.

Standing beside the game is not a role. Billy now responds to every action, which gives her one reason to be on screen.

WHAT I WOULD DO DIFFERENTLY NOW

Build the clickable version first.

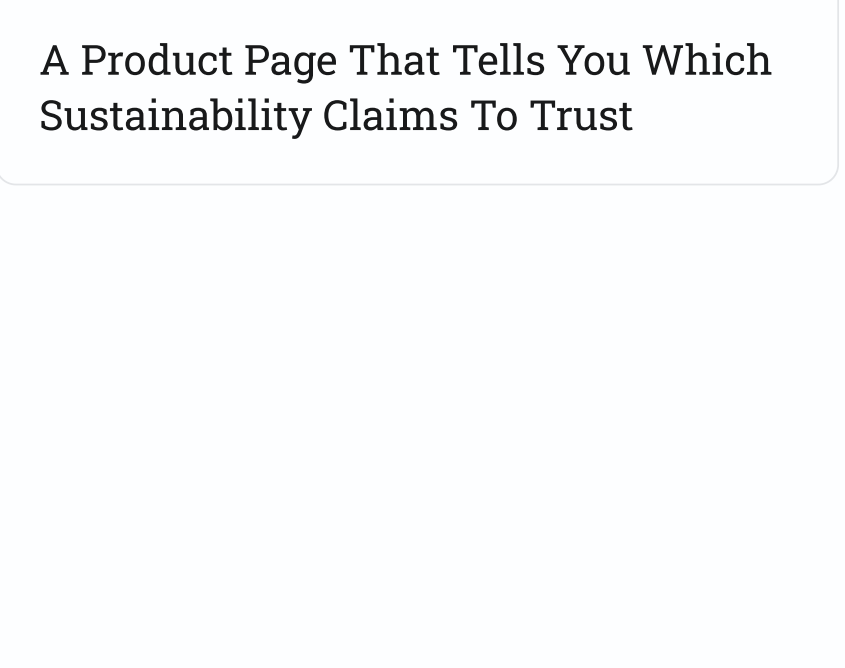
I tested on paper because a digital prototype was expensive to make. That reason has mostly gone, and the test I actually needed was always the digital one.

Let children write the item list.

A short survey and a pass over the answers would give me the categories the game should use, instead of the ones I assumed.

> Original project by COllab Studio. Visual language, style guide, illustrations and branding by Sharanya. Story and narrative by Kaitlyn Carney and Dvija Patel. Book layouts by Sam Collins. Interactions and pop-ups by Kaitlyn Carney. Website mini games by Dvija Patel. The four diagrams on this page were redrawn by me from the team's process book, content unchanged.

ALSO CHECKOUT



GREENERA UI/UX DESIGN 2025

A Product Page That Tells You Which Sustainability Claims To Trust

THE FAMILY TABLE UI/UX DESIGN 2025

A Weekly Family Meal Everyone Wants To Be Part Of

Intro
TL;DR
Brief
Challenge
Solution
The Problem
Initial Findings
The System
Design Process
Turning Point
Goal
The Business Case
Outcome
Reflection